GradWise:Adaptive AI Learning Assistant for College Students

ASE Project Review-1

Date& Time

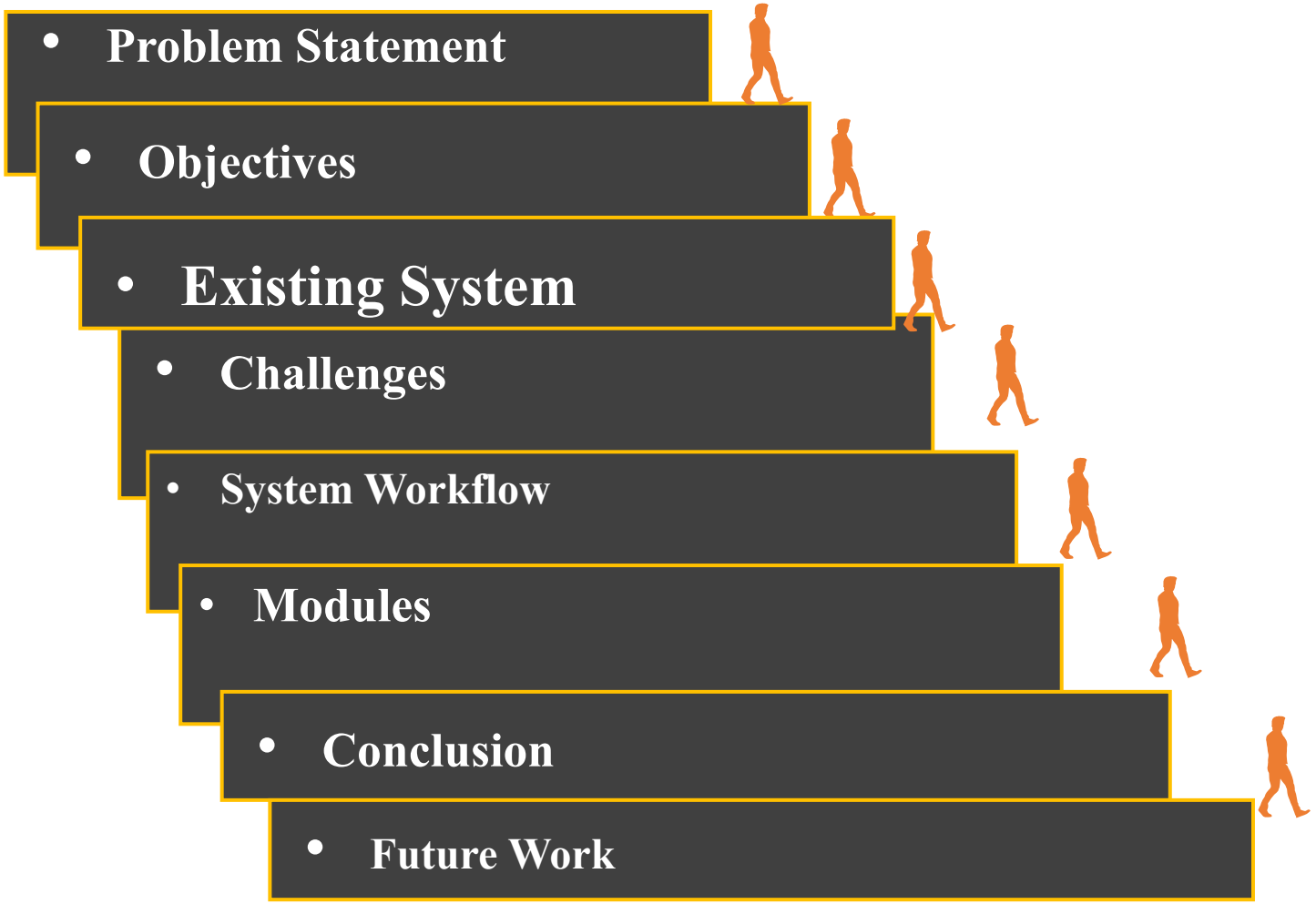
Under the Guidance of

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Outline of Project Work



Problem Statement

1.Problem Statement

College students have different learning abilities, academic needs, and study patterns. Existing learning systems often fail to provide continuous, personalized support, making it difficult for students to identify weak subjects, plan their studies, and improve academic performance.

2. Problems with the Current Approach

One-size-fits-all learning: Limited adaptation to individual learning needs.
Manual progress tracking: Students struggle to identify weak subjects and learning gaps.
Static study plans: Study schedules may not adjust to changing performance.
Limited personalized pra**ctice:** Quizzes and explanations may not match individual knowledge gaps.

3. Our Proposed Solution – GradWise

GradWise is an AI-powered adaptive learning assistant that analyzes student performance, identifies learning gaps, generates personalized quizzes and explanations, and recommends adaptive study plans to support continuous academic improvement.

Objectives

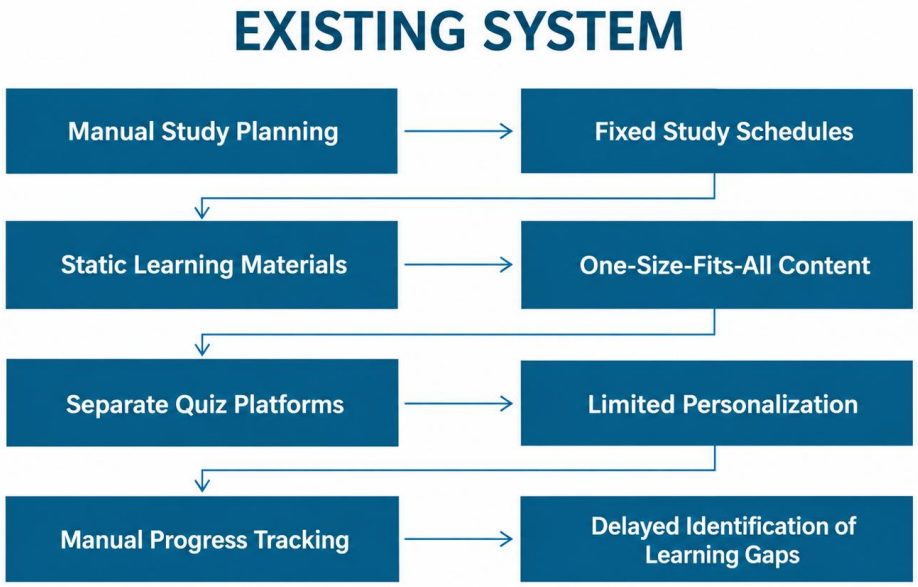
Objectives

Personalized Learning: Adapt learning support to each student's needs and performance.
Identify Learning Gaps: Analyze quiz results and academic progress to identify weak areas.
Adaptive Quiz Generation: Generate AI-powered quizzes based on students' learning needs.
Smart Study Planning: Recommend personalized study plans and track study progress.
Continuous Improvement: Update learning recommendations based on student performance and interactions.

Our Goal

To develop an AI-powered adaptive learning assistant that helps college students learn more effectively through personalized quizzes, intelligent study plans, progress tracking, and AI-generated explanations—supporting continuous academic improvement.

Existing System & Limitations



- **Limitations**

➢ **Limited Personalization:** Learning materials and quizzes may not adapt to individual students' learning needs.

➢ **Difficulty Identifying Weak Areas:** Students often struggle to identify knowledge gaps and track subject-wise performance.

➢ **Static Study Plans:** Study schedules do not automatically adjust to students' progress or changing learning needs.

➢ **Generic Feedback and Support:** Explanations and recommendations may not address each student's specific difficulties.

Proposed System

GradWise is an AI-powered adaptive learning assistant designed to help college students improve their academic performance through personalized learning support.It integrates AI-generated explanations, adaptive quizzes, personalized study plans, and progress tracking into a single platform.

- **AI-Powered Learning**: Provides personalized explanations based on students' learning needs.
- **Adaptive Quiz Generation**: Generates quizzes to practice concepts and identify learning gaps.
- **Personalized Study Plans**: Recommends study schedules based on learning progress.
- **Progress Tracking**: Tracks completed study sessions, quiz history, and academic progress.

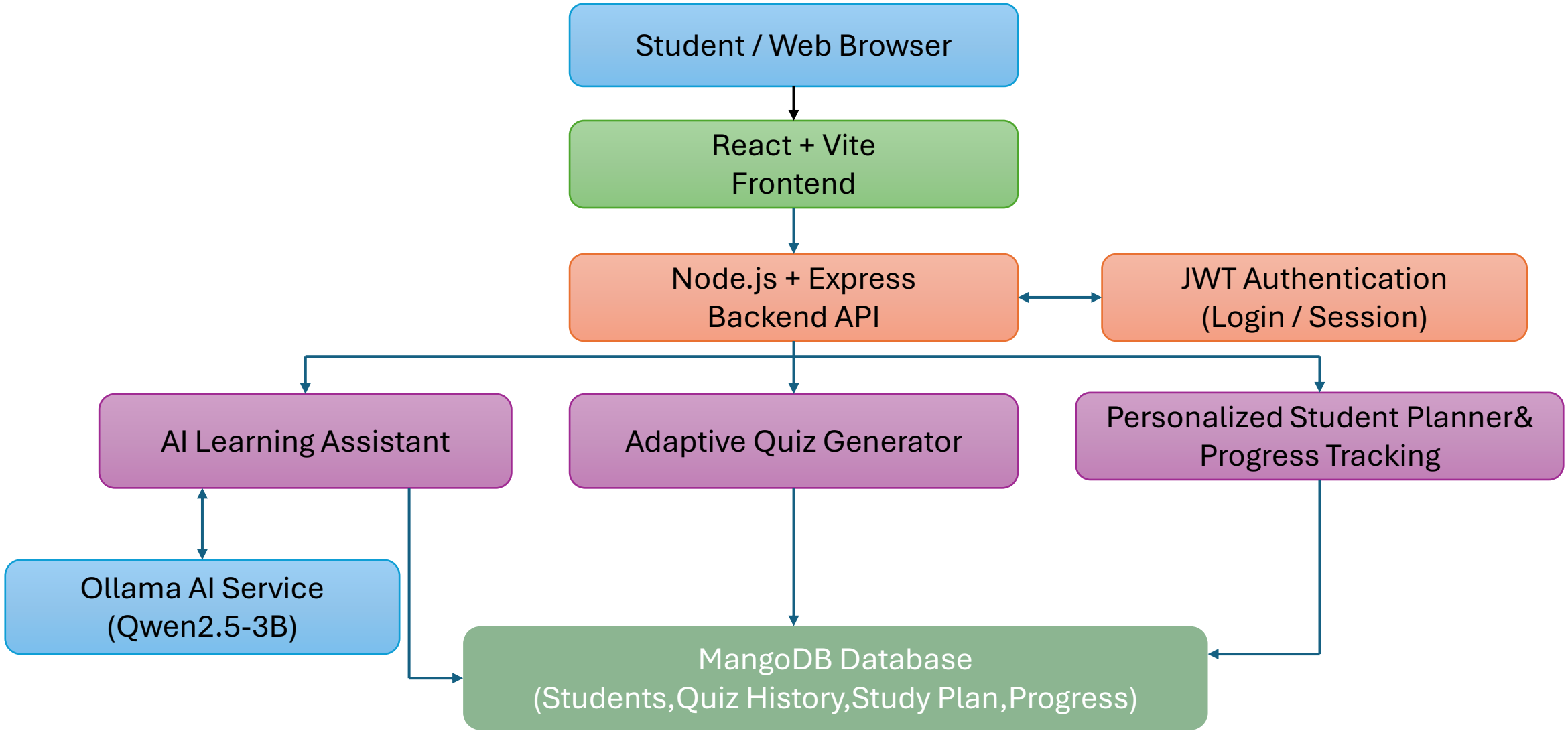
Key Benefits

1. Personalized Experience: Supports individual learning requirements.
2. Improved Learning: Helps students identify and focus on weak subjects.
3. Better Time Management: Organizes study activities through structured plans.
4. Continuous Improvement: Helps students monitor progress and improve their learning habits.

Technology Used

Layer	Technology
Frontend	React.js + Vite
UI	CSS + Lucide React
Backend	Node.js + Express.js
Database	MangoDB
Data Modeling	Mongoose
Authentication	BCrypt + JWT
API	REST API
AI Integration	Ollama
AI Model	Owen 2.5(3B)
Development Tools	Visual Studio Code
Package Manager	npm
Version Control	Git + GitHub

System Workflow



Modules

Module	Status	Description
User Authentication	✅ Completed	Secure student login and JWT-based authentication implemented.
Student Dashboard	✅ Completed	Displays student information, study progress, and learning activities.
AI Learning Assistant	✅ Completed	Provides an AI chat interface for learning support and explanations.
Adaptive Quiz Generator	✅ Completed	Generates quizzes to help students practice concepts and assess their knowledge.
Quiz History	✅ Completed	Stores quiz attempts and allows students to review previous quiz activity.
Personalized Study Planner	✅ Completed	Creates personalized study plans to organize learning activities.
Study Progress Tracking	✅ Completed	Tracks completed study sessions and displays learning progress.
AI Model Integration	✅ Completed	Integrates the local Ollama service using the Qwen 2.5 (3B) model.

Implementation Progress

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Frontend

- ✅ **React + Vite Setup** – Provides a modern and fast frontend development environment.
- ✅ **Login Interface** – Allows students to securely access their accounts using authentication.
- ✅ **Student Dashboard** – Displays student information, study activities, and learning progress.
- ✅ **AI Learning Assistant Interface** – Provides a chat interface for AI-powered learning support and explanations.
- ✅ **Adaptive Quiz Interface** – Allows students to generate and practice quizzes.
- ✅ **Study Planner Interface** – Helps students organize study activities and view personalized study plans.
- ✅ **Progress Tracking Interface** – Displays completed study sessions and tracks learning progress.

Implementation Progress

Backend

- ✅ **Node.js + Express Setup** – Provides REST APIs and core backend services.
- ✅ **MongoDB Integration** – Stores student accounts, quiz history, study plans, and learning data.
- ✅ **Student Management** – Manages student records and account information.
- ✅ **Authentication APIs** – Handles student login and authentication requests.
- ✅ **Bcrypt Password Hashing** – Protects student passwords using secure hashing.
- ✅ **JWT Authentication** – Authenticates users through JSON Web Tokens.
- ✅ **Protected APIs** – Restricts access to authorized users through authentication middleware.
- ✅ **Student Profile API** – Retrieves information about the currently logged-in student.
- ✅ **Quiz APIs** – Handles quiz generation, quiz attempts, and quiz history.
- ✅ **Study Planner APIs** – Manages personalized study plans and study progress.

Implementation Progress

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- **Learning Data Management**

✅ **Quiz History** – Stores students' quiz attempts and results for future reference.

✅ **Student-Specific Data** – Maintains learning information associated with individual student accounts.

✅ **Study Plan Management** – Stores and manages personalized study plans.

✅ **Progress Tracking** – Records completed study sessions and displays learning progress.

- **AI Service**

✅ **Ollama AI Service** – Uses the local Ollama service to run the Qwen 2.5 (3B) language model.

✅ **Backend Integration** – Connects the Node.js and Express backend with the AI service for learning-related requests.

✅ **AI Learning Assistant** – Provides a chat interface for AI-powered explanations and learning support.

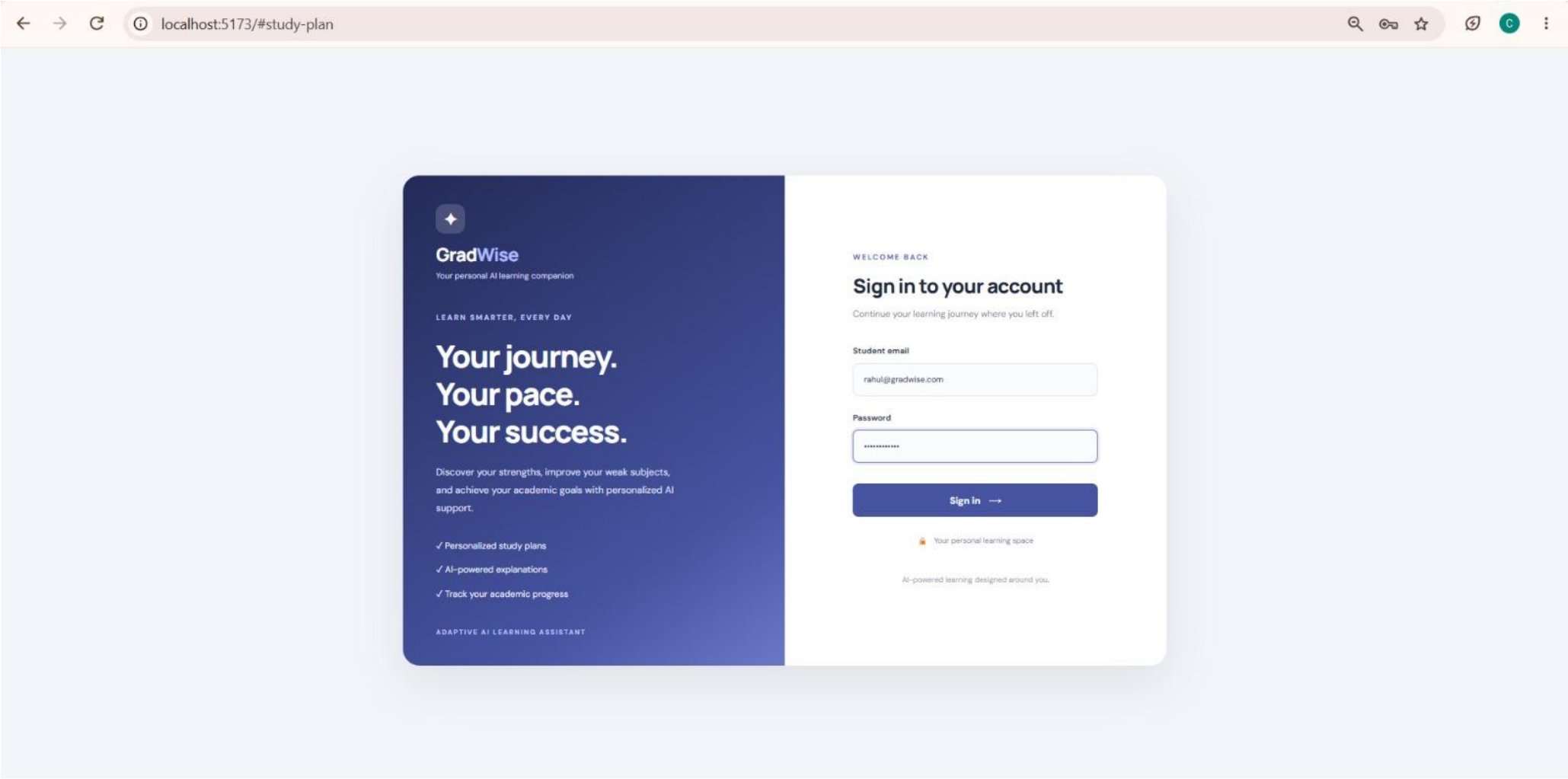
✅ **Adaptive Quiz Generation** – Supports quiz generation to help students practice and assess their knowledge.

⏳ **Advanced AI Personalization** – Future enhancement to tailor recommendations more deeply to individual learning patterns.

Working Module Screenshots

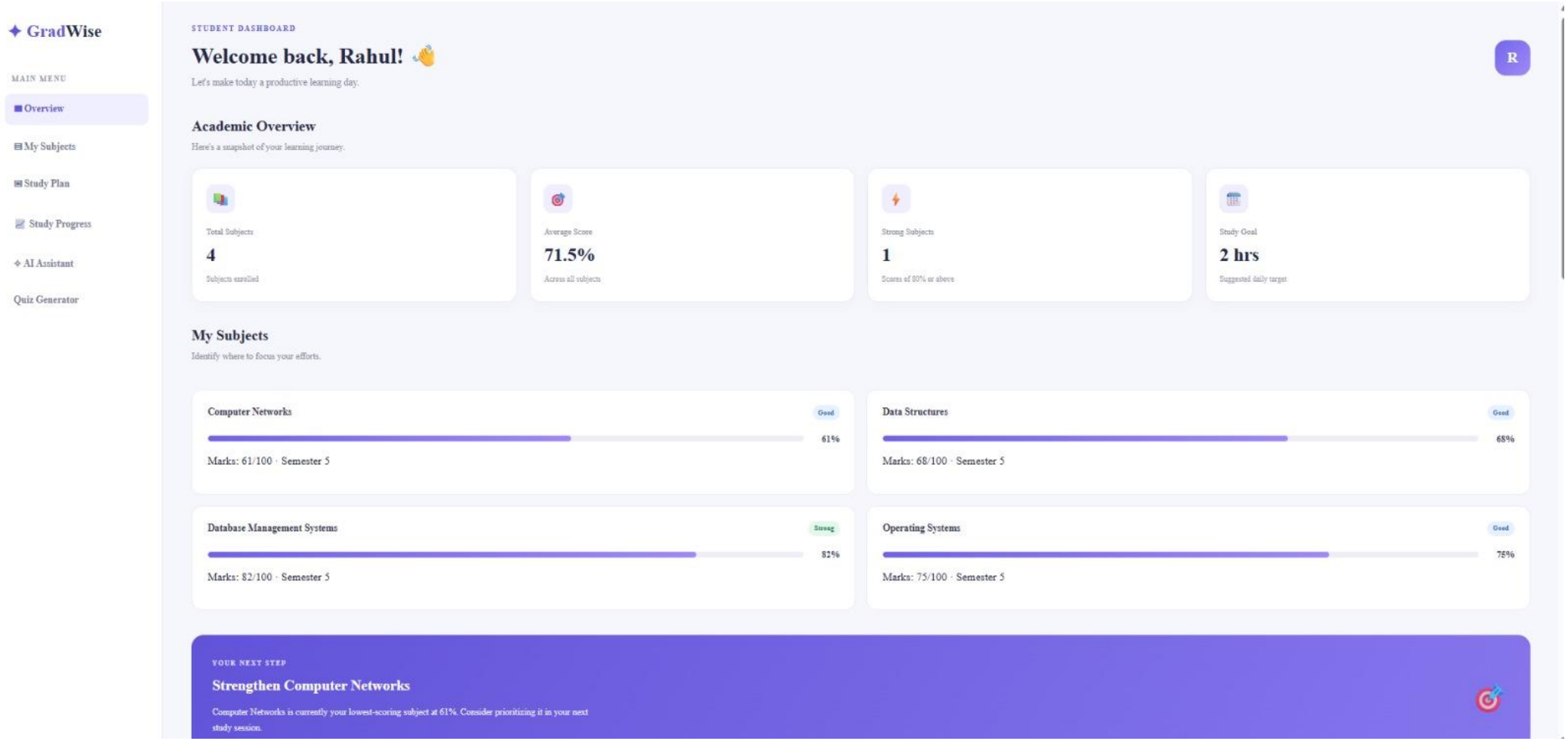
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Screenshot 1:Login Page



Working Module Screenshots

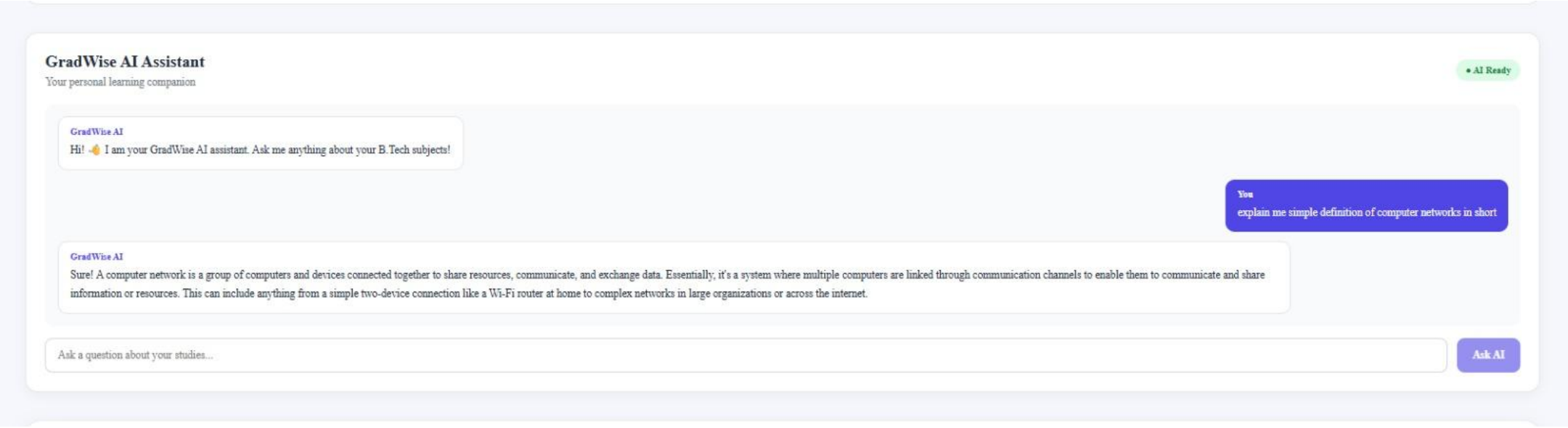
Screenshot 2:Dashboard



Working Module Screenshots

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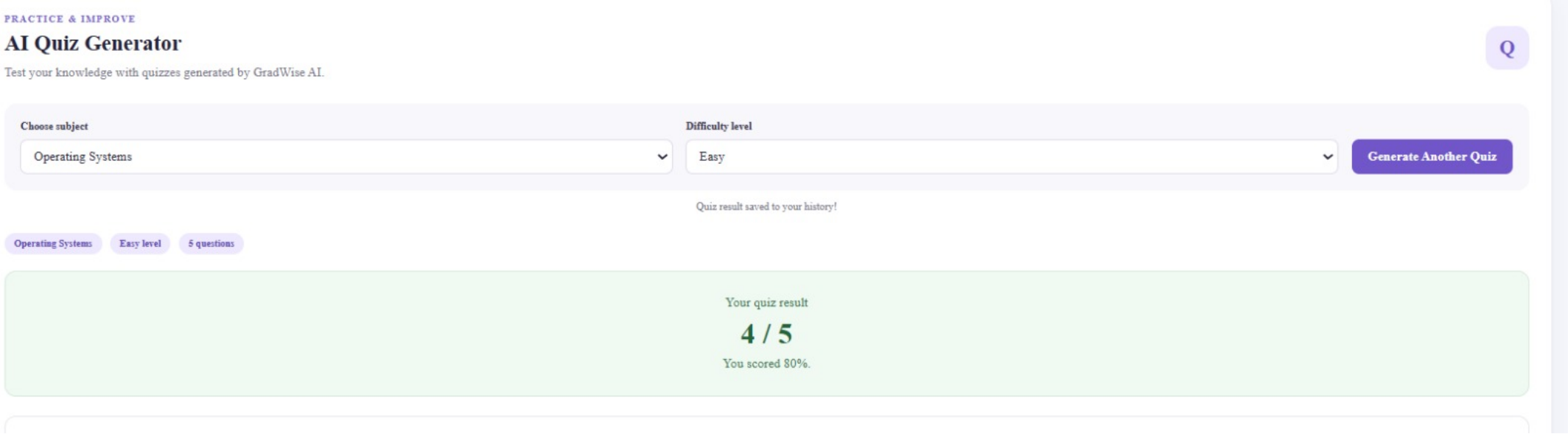
Screenshot 3:AI Assistant



Working Module Screenshots

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Screenshot 4:AI Quiz Generator



Which of the following is a benefit of using a virtual memory system?

- A It increases the physical memory available to the system
- B It reduces the need for a file system
- C It allows for larger address spaces than physical memory
- D It simplifies the memory management system

QUESTION 3 OF 5

What is the primary function of a deadlock avoidance algorithm?

- A To prevent deadlocks from occurring
- B To detect deadlocks after they have occurred
- C To resolve deadlocks by terminating processes
- D To prevent processes from entering a deadlock state

QUESTION 4 OF 5

Which scheduling algorithm is known for having the longest average waiting time for a process?

- A First-Come, First-Served (FCFS)
- B Shortest Job First (SJF)
- C Round Robin (RR)
- D Priority Scheduling

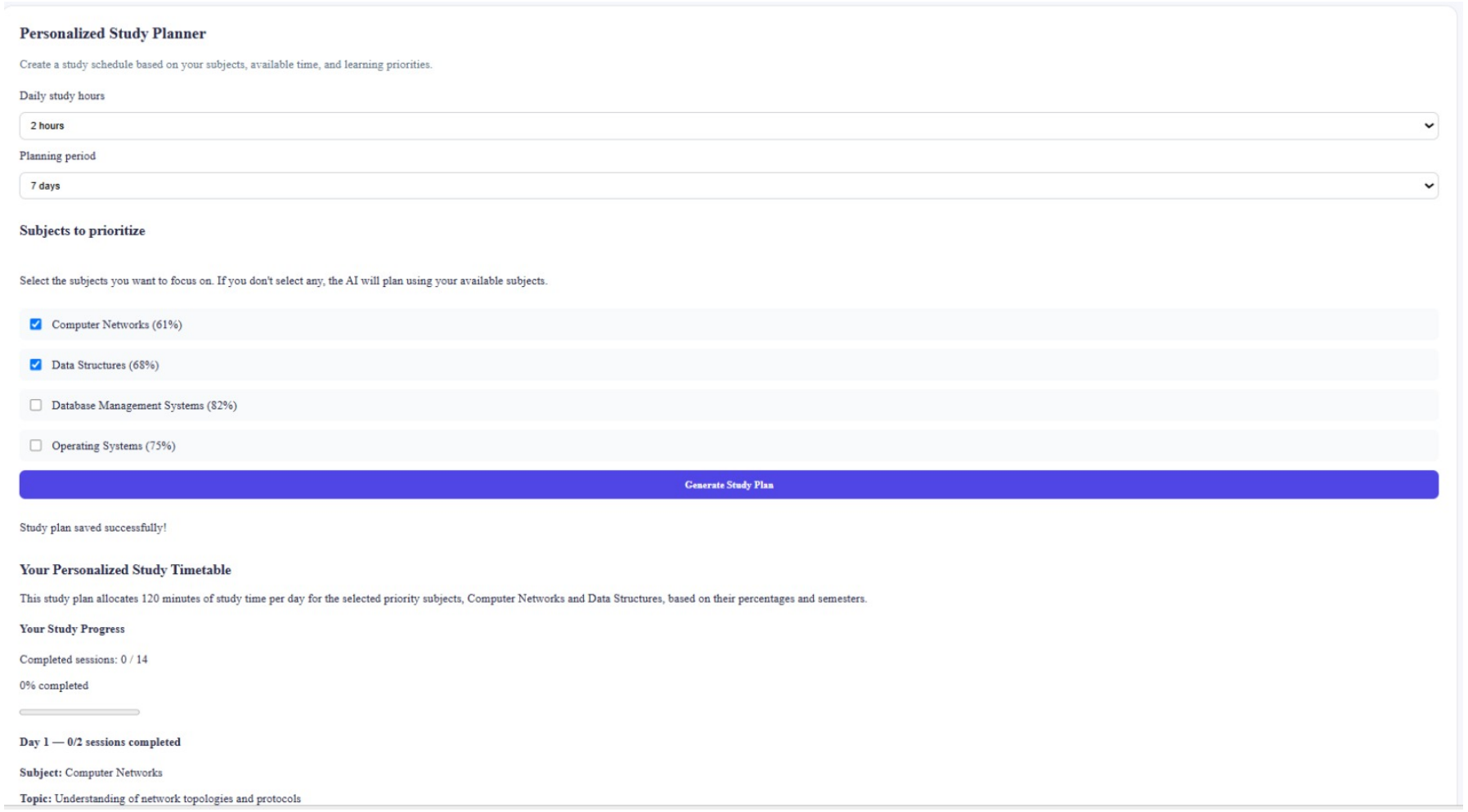
Working Module Screenshots

Screenshot 5:AI Quiz History



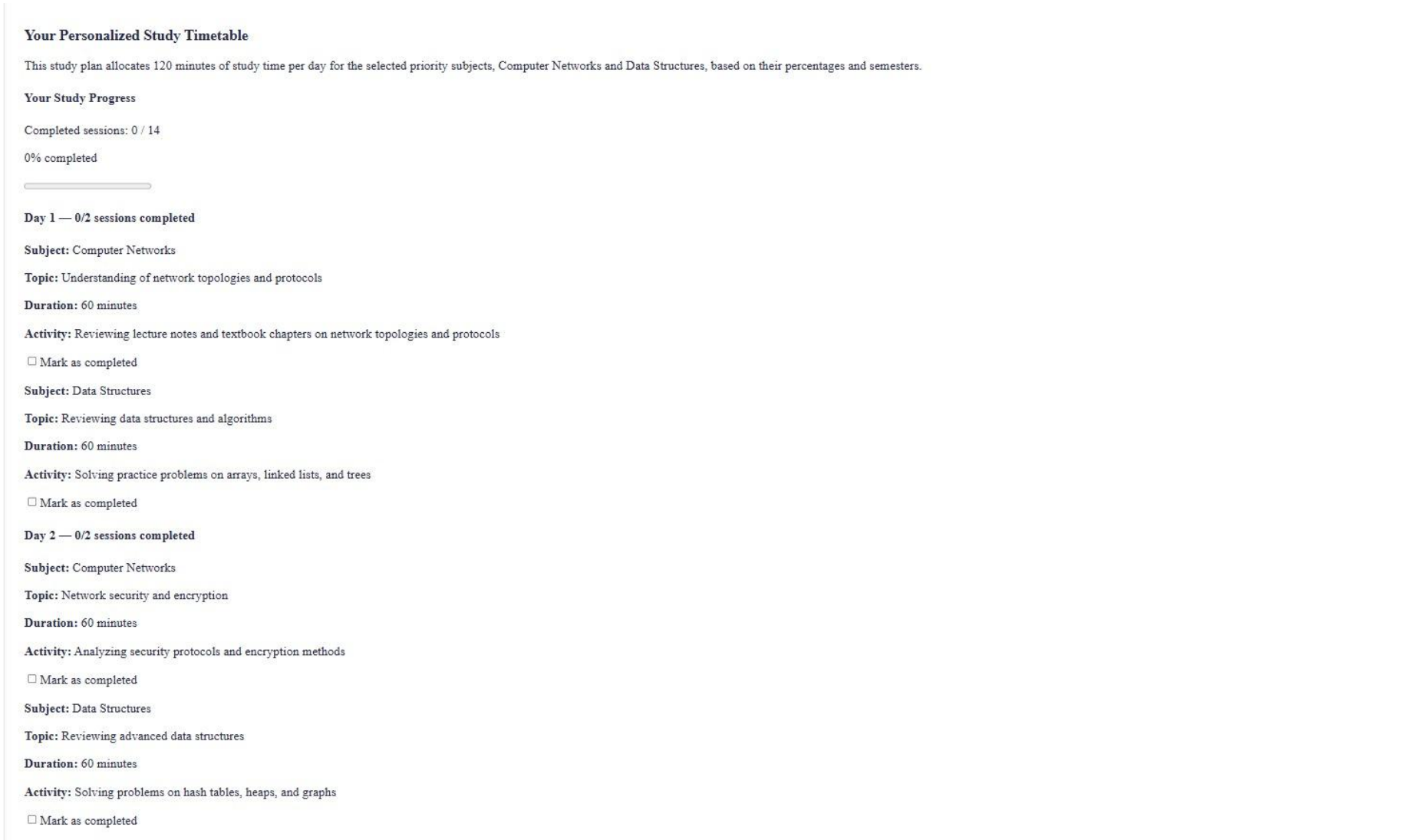
Working Module Screenshots

Screenshot 6:Personalized Study Planner



Working Module Screenshots

Screenshot 6:Personalized Study TimeTable



Your Personalized Study Timetable

This study plan allocates 120 minutes of study time per day for the selected priority subjects, Computer Networks and Data Structures, based on their percentages and semesters.

Your Study Progress

Completed sessions: 0 / 14

0% completed

Day 1 — 0/2 sessions completed

Subject: Computer Networks

Topic: Understanding of network topologies and protocols

Duration: 60 minutes

Activity: Reviewing lecture notes and textbook chapters on network topologies and protocols

☐ Mark as completed

Subject: Data Structures

Topic: Reviewing data structures and algorithms

Duration: 60 minutes

Activity: Solving practice problems on arrays, linked lists, and trees

☐ Mark as completed

Day 2 — 0/2 sessions completed

Subject: Computer Networks

Topic: Network security and encryption

Duration: 60 minutes

Activity: Analyzing security protocols and encryption methods

☐ Mark as completed

Subject: Data Structures

Topic: Reviewing advanced data structures

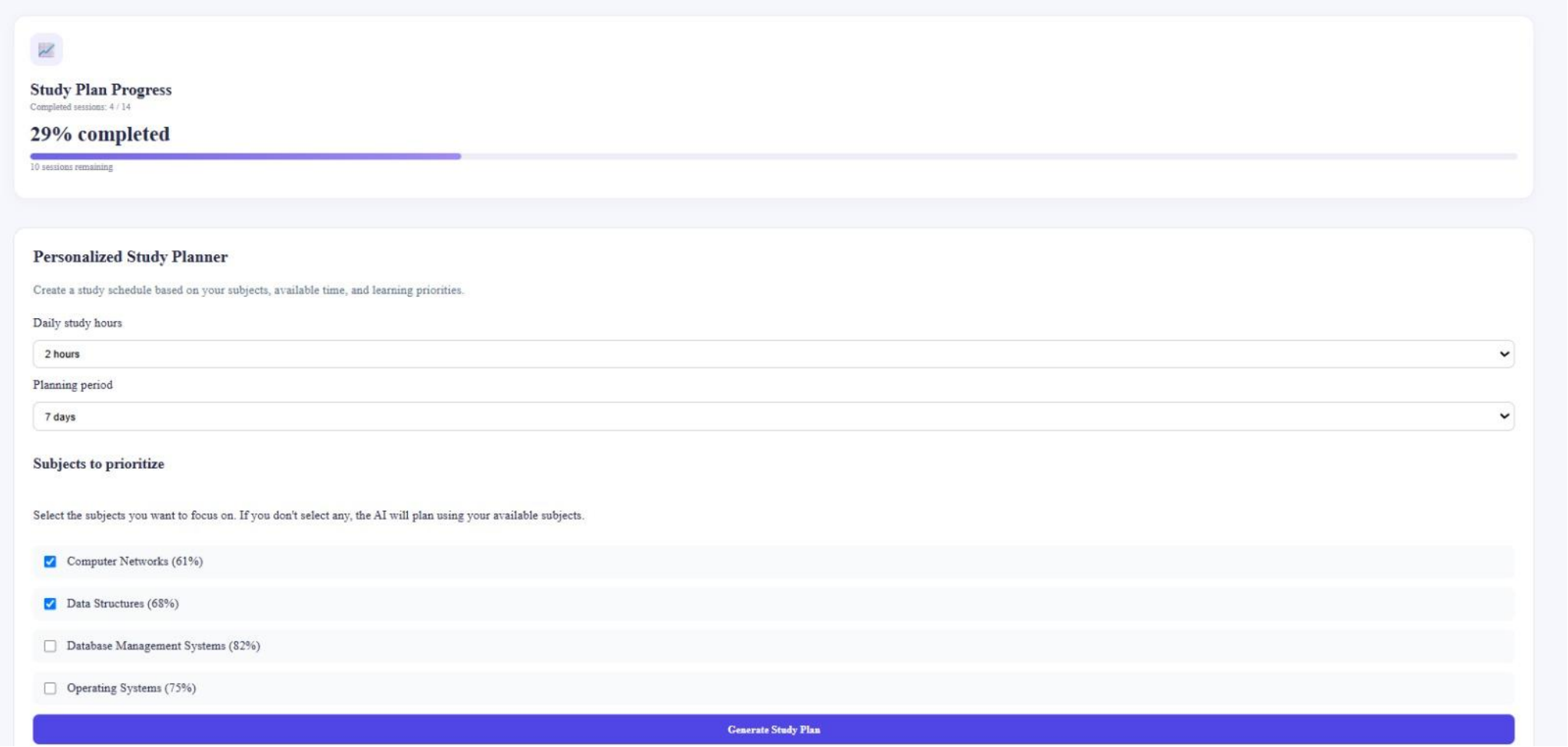
Duration: 60 minutes

Activity: Solving problems on hash tables, heaps, and graphs

☐ Mark as completed

Working Module Screenshots

Screenshot 6:Study Plan Progress



Conclusion

Conclusion

GradWise has successfully established the core foundation of an AI-powered adaptive learning platform for college students. Its major learning modules have been implemented and integrated, providing a foundation for personalized learning support, intelligent study planning, and continuous academic improvement.

Currently Working

✅ **Student Authentication** – Students can securely log in to their accounts.
✅ **JWT Authentication** – Token-based authentication protects student sessions and authorized APIs.
✅ **Personalized Dashboard** – Displays student information, study activities, and learning progress.
✅ **AI Learning Assistant** – Provides a chat interface for AI-powered learning support and explanations.
✅ **Adaptive Quiz Generator** – Helps students practice concepts through generated quizzes.
✅ **Quiz History** – Stores quiz attempts and allows students to review previous quiz activity.
✅ **Personalized Study Planner** – Helps students organize their study activities and learning goals.
✅ **Study Progress Tracking** – Tracks completed study sessions and displays learning progress.
✅ **MongoDB Integration** – Stores student accounts, quiz history, and study-related information.

Future Scope

1. AI Learning Assistant
More intelligent and personalized learning assistance
Natural-language interaction for student queries
Integration with advanced AI models for better explanations

2. Advanced Adaptive Quizzes
Dynamic quiz generation based on student performance
Automatic difficulty adjustment based on learning progress
Personalized feedback and recommendations

3. AI-Powered Learning Materials
Upload and analyze study materials and documents
Document-based question answering
Retrieval-Augmented Generation (RAG) for context-based answers

4. Advanced AI Personalization
Personalized learning recommendations based on student performance
Identification of weak subjects and learning gaps
Machine learning-based academic performance prediction

5. Smart Study Planning
Automatically updated study plans based on quiz results
Intelligent scheduling of study sessions and academic goals
Recommendations to improve time management and learning outcomes

6. Advanced Security
Improved authentication and access control
Enhanced protection of student information
Additional privacy and data security mechanisms

Final Statement

- GradWise currently provides a functional foundation for AI-powered adaptive learning, combining student authentication, personalized dashboards, AI-assisted learning, quiz generation, study planning, and progress tracking. Future enhancements will focus on advanced AI personalization, document-based learning, machine learning-based performance prediction, and scalable deployment to deliver a smarter and more effective learning experience for students.